

Aitchison-Loss Training with Geospatial Embeddings Sharpens Compositional Land-Cover Maps

Ayato Kanno¹
Narumasa Tsutsumida^{1,2}

¹ Graduate School of Science and Engineering, Saitama University, Japan

² Space Data Frontiers Research Center, Fujitsu Research, Fujitsu Limited, Japan

FOSS4G Hiroshima 2026

September 3, 2026

Importance of Land-Cover Mapping

Land-cover maps derived from satellite imagery serve as fundamental data for decision-making in various fields.

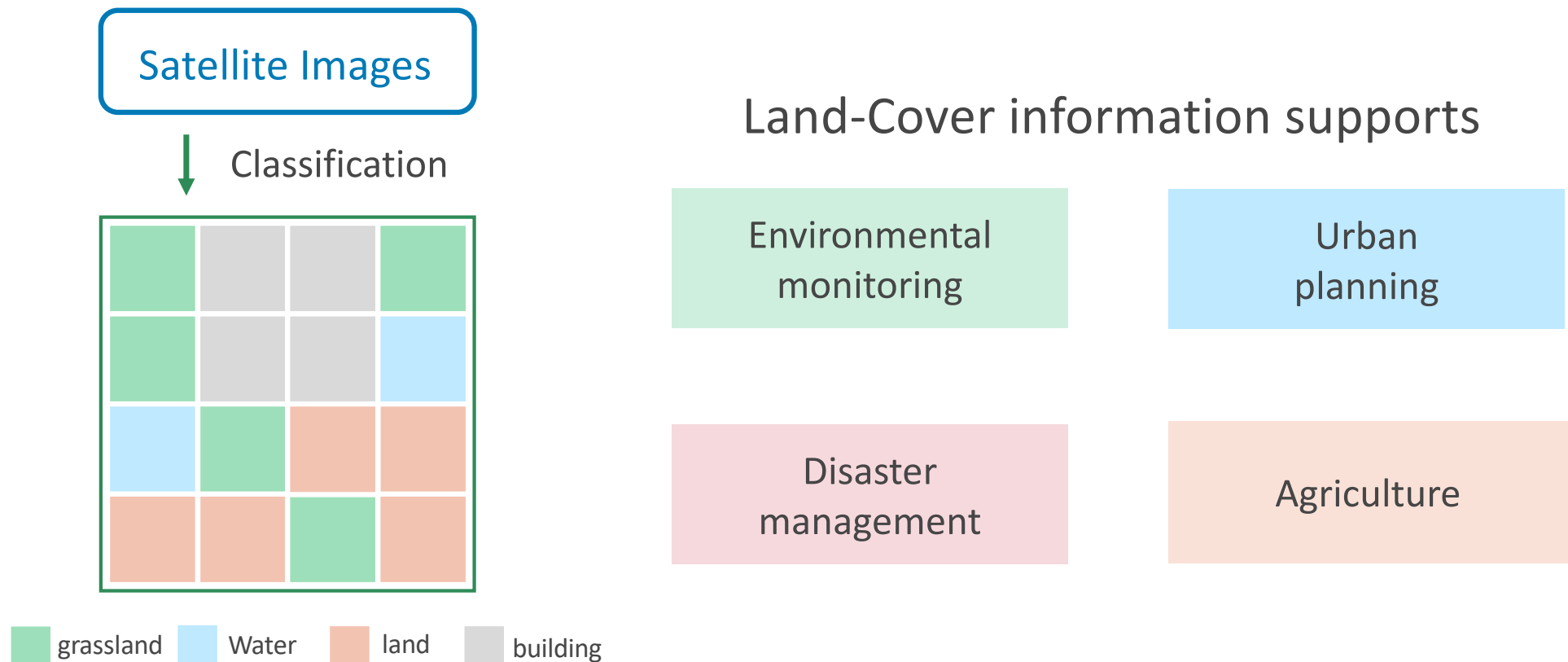


Illustration of a Land-Cover Mapping

Mixed-Pixel Problem

Limited Spatial Resolution of Satellite Imagery

Landsat-8 : 30 m Sentinel-2 : 10 m

Complex region

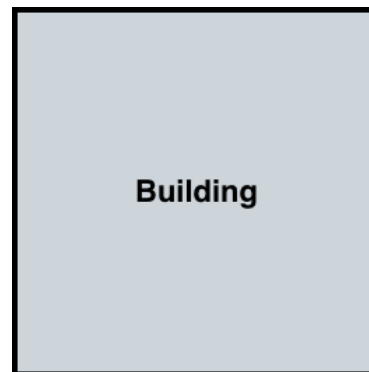
→ A single pixel may contain multiple land-cover classes



Hard Classification

One class per pixel

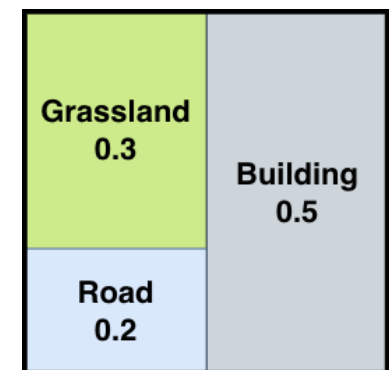
Minor classes discarded



Fractional Classification

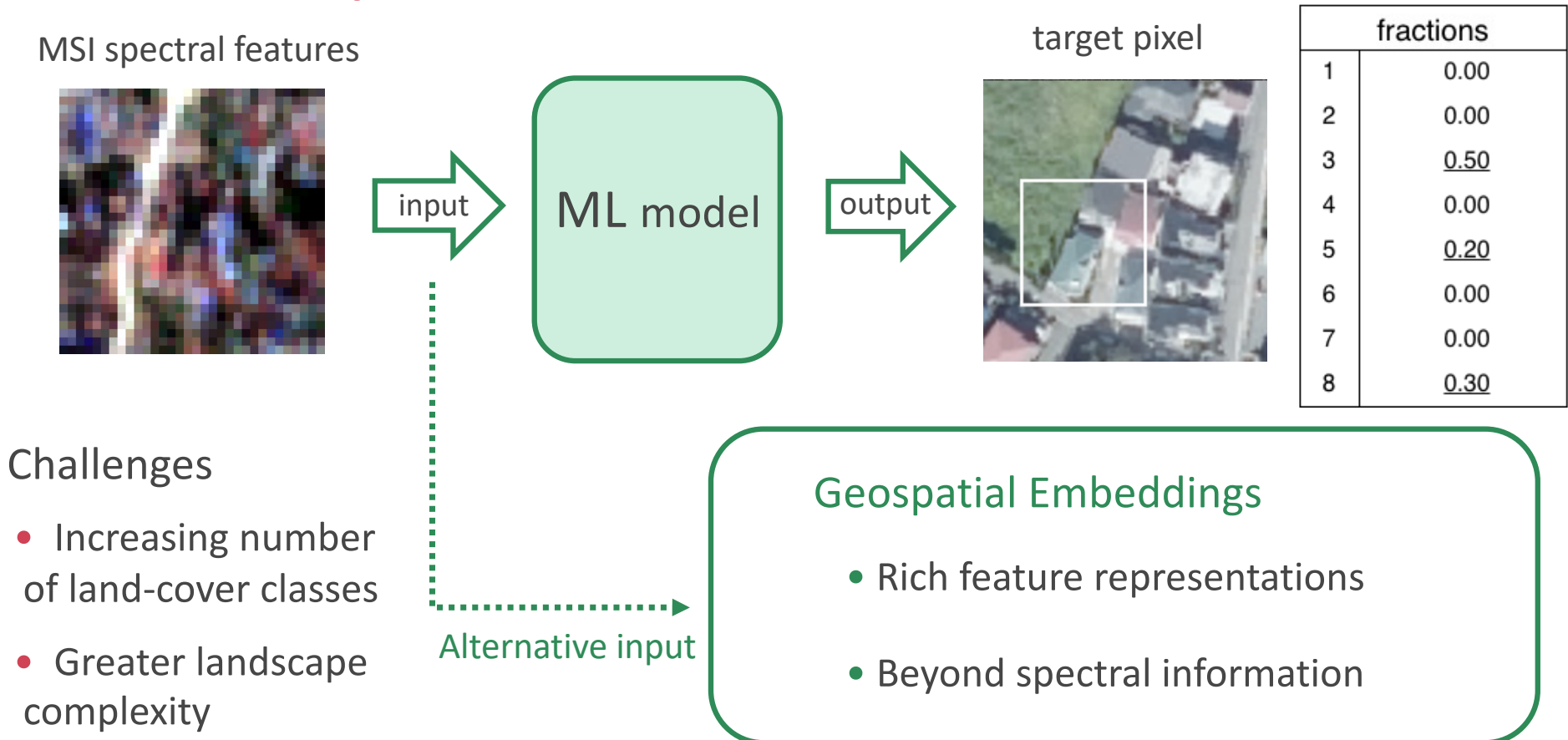
Class proportions

Minor classes preserved



Fraction = proportion of each class within a pixel

Limitation of Spectral Features



Need both richer feature representations and appropriate ML models

Limitations of Standard Regression Losses to train a ML model

Land-cover class fractions are compositional data

- Non-negative
Each class fraction is ≥ 0
- Sum-to-one
The fractions sum to 1 (100%)



fractions	
1	0.00
2	0.00
3	<u>0.50</u>
4	0.00
5	<u>0.20</u>
6	0.00
7	0.00
8	<u>0.30</u>

Components have relative relationships

Standard regression losses ignore these relationships

e.g., MAE, RMSE

Geospatial Embeddings

Provide richer geospatial information



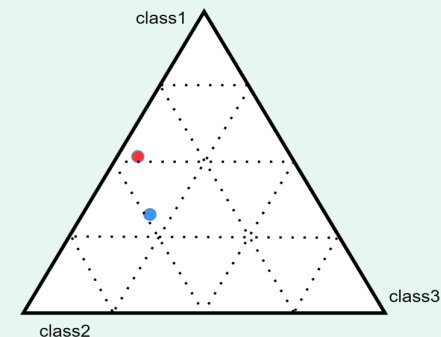
Embedding Vector

- High-dimensional representations
- Capture diverse geospatial context



Compositional Loss

Respect compositional properties



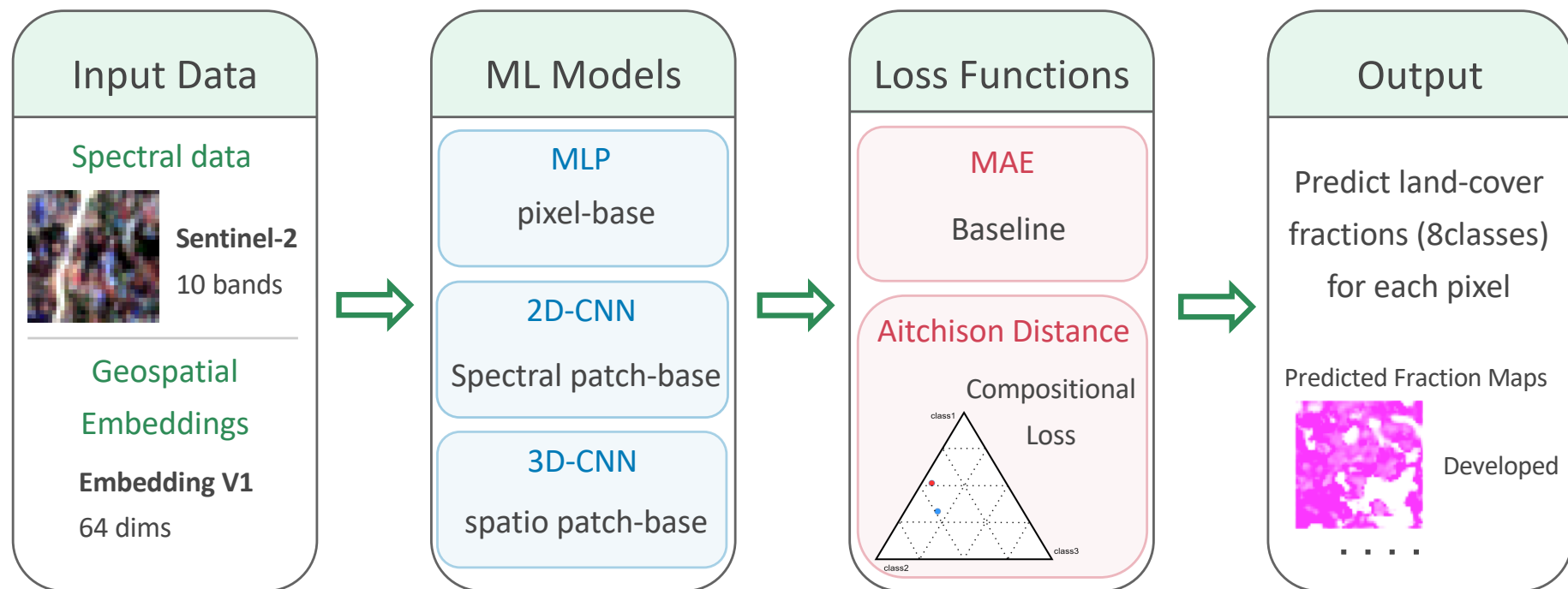
- Suitable for Land-Cover fractions
- Account for relative relationships



Improve Fractional Land-Cover Mapping

Method Overview

Compare different **input features** and **models** for compositional land-cover estimation using **compositional loss functions**



Input Data

Sentinel-2 Multispectral Imagery

- Spatial resolution: 10-20 m
Resampled to 10 m
- 10 spectral bands
(RGB, NIR, Red Edge, SWIR)
- Annual median composite (2020)
- Cloud and shadow masking



Widely used for land-cover mapping (Chaves et al., 2020)

Input Data

Google Satellite Embedding V1

- 64-dimensional embedding vector
One vector for each 10 m grid cell
- Beyond spectral reflectance
Learned from multiple Earth observation data sources
Encodes spectral, spatial, temporal, and semantic information

Generated by AlphaEarth Foundations v2.1

Annual embedding (2020/04/01–2021/04/01)

Previous work demonstrated the effectiveness of AlphaEarth embeddings for categorical classification (Khan & Ahmad, 2025)

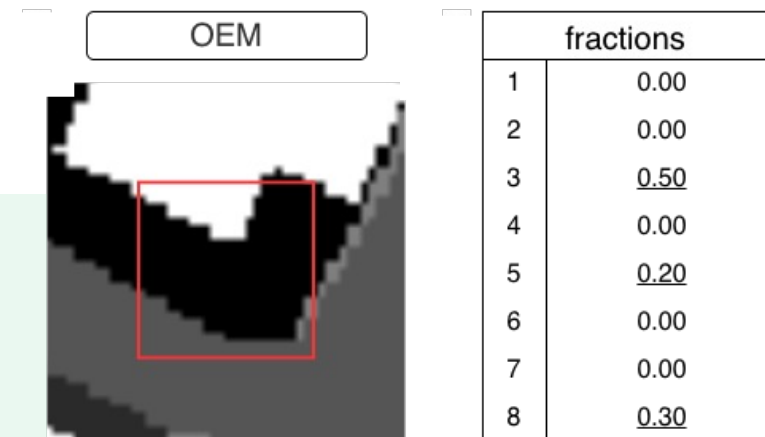
Khan, M. A., & Ahmad, A. (2025). *Evaluating AlphaEarth foundation embeddings for pixel- and object-based land cover classification in Google Earth Engine*.

Reference Data

OpenEarthMap (OEM)

- Spatial resolution: 0.25–0.5 m
- 97 regions in 44 countries
- 2,300 annotated images

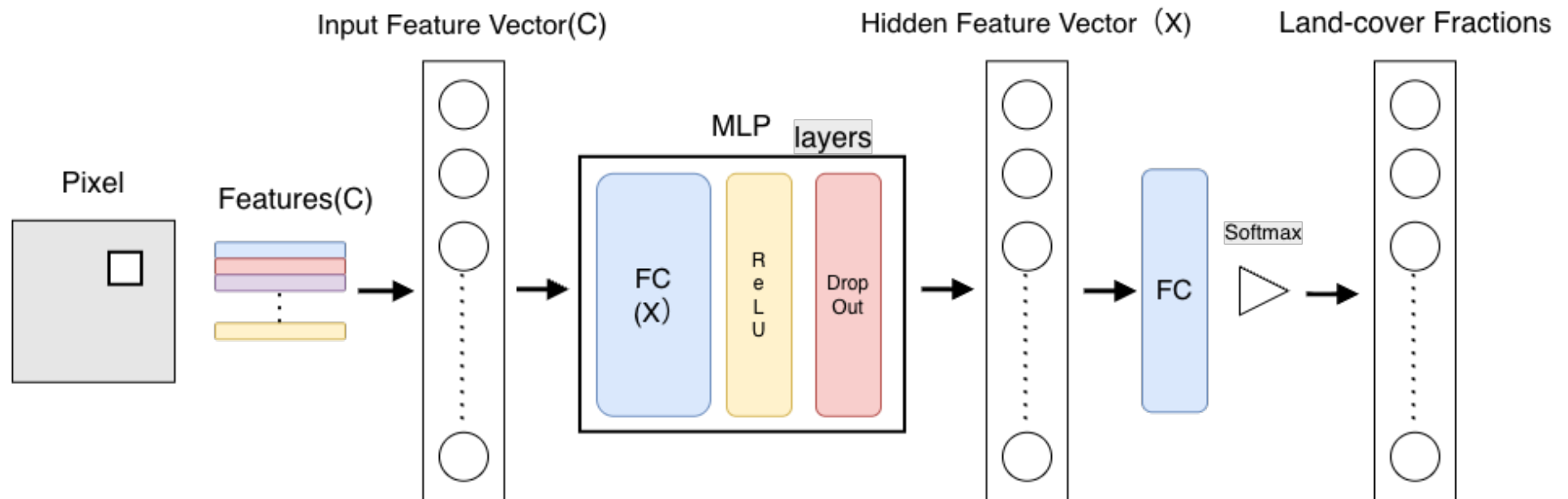
Bare	Grass	Developed	Road
Forest	Water	Cropland	Building



Aggregated to the Sentinel-2 (10m) grid

→ Land-cover fractions

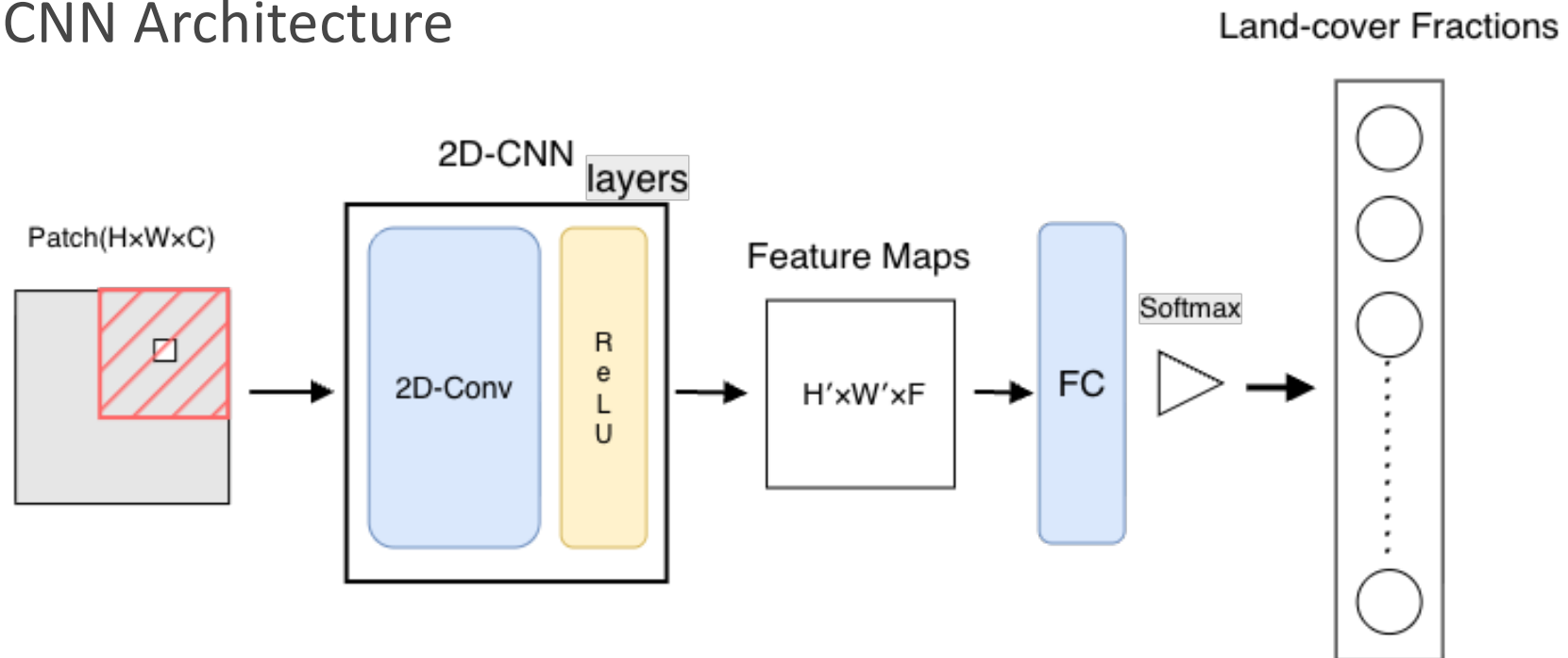
MLP Architecture



- 3 fully connected layers
- ReLU activation
- Dropout (20%)
- Softmax output

- Processes each pixel independently
- Simple architecture

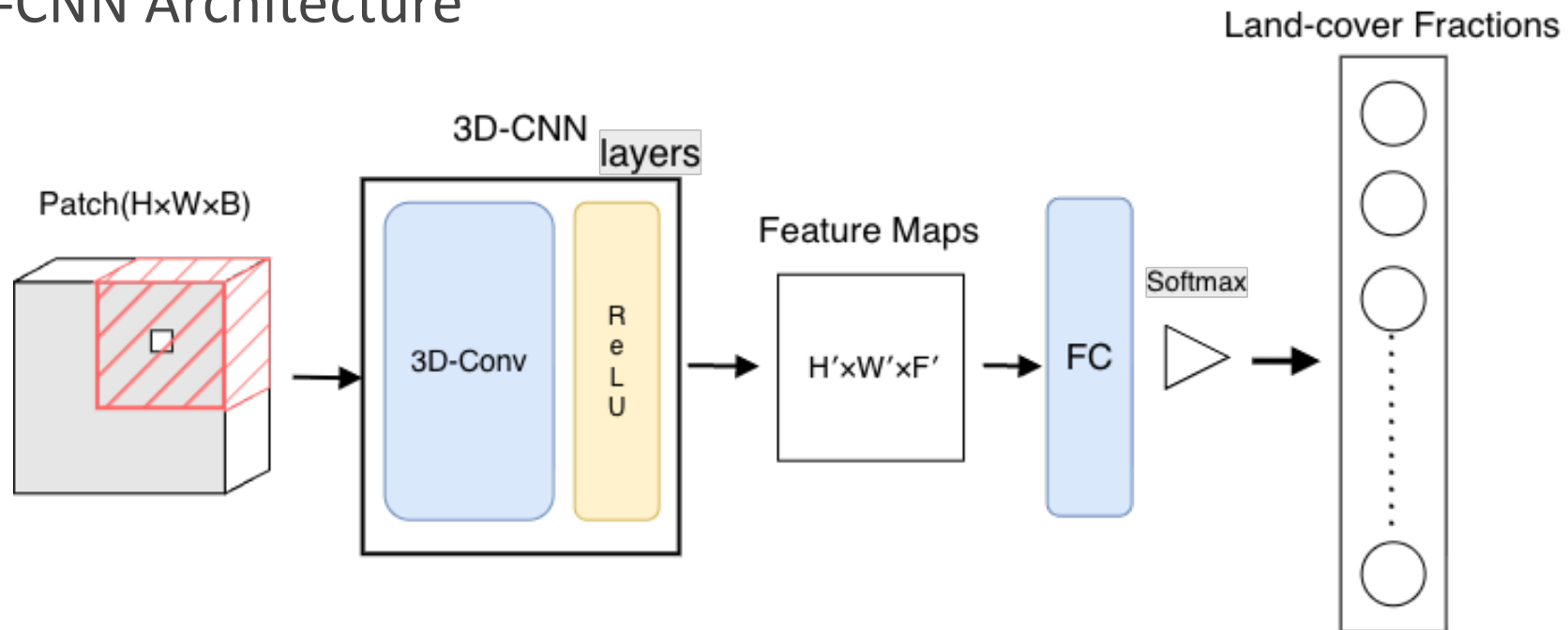
2D-CNN Architecture



- 4 convolutional layers
- ReLU activation
- Batch normalization
- Softmax output

- Learns local spatial features
- Moderately complex architecture

3D-CNN Architecture



- 4 convolutional layers
- ReLU activation
- Batch normalization
- Softmax output

- Learns joint spatial–spectral features
- More complex architecture

Regression Loss

Mean Absolute Error (MAE)

$$\text{MAE} = \frac{1}{N} \sum_{n=1}^N \left(\frac{1}{D} \sum_{i=1}^D |y_{n,i} - \hat{y}_{n,i}| \right)$$

- Measures absolute differences
- Treats all classes equally

1. Compute the **absolute difference for each class**
2. **Average** the differences

Does not consider compositional properties

Compositional Loss

Aitchison Distance

$$d_A(\mathbf{y}, \hat{\mathbf{y}}) = \sqrt{\sum_{i=1}^D \left(\ln \frac{y_i}{g(\mathbf{y})} - \ln \frac{\hat{y}_i}{g(\hat{\mathbf{y}})} \right)^2}$$

- Penalizes errors near zero
- Measures relative differences

1. Apply the **centered log-ratio (CLR)** transformation
2. Compute the **Euclidean distance** in the CLR space

**Accounts for relative relationships among
compositional variables**

① Effect of Input Features

- Sentinel-2 spectral data
- Google Satellite Embedding V1

② Effect of Loss Function

- MAE
- Aitchison distance

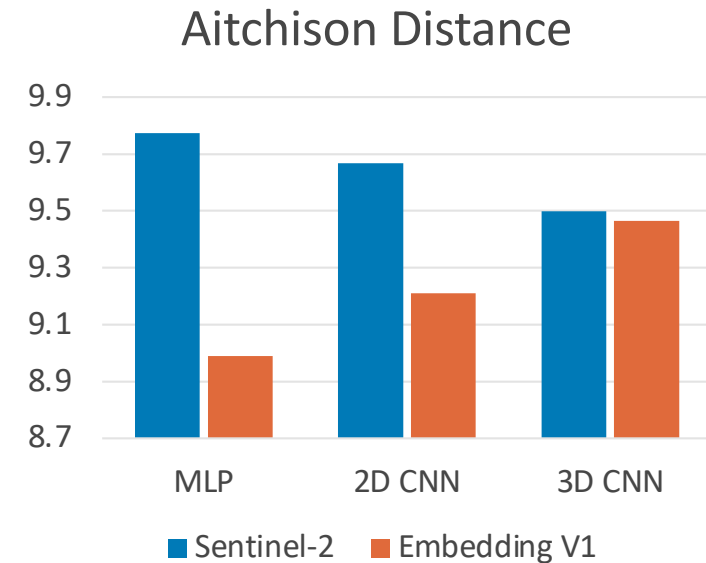
Evaluation

- Quantitative evaluation using MAE, RMSE, and Aitchison distance
- Qualitative comparison of land-cover fraction maps

① Effect of Input Feature

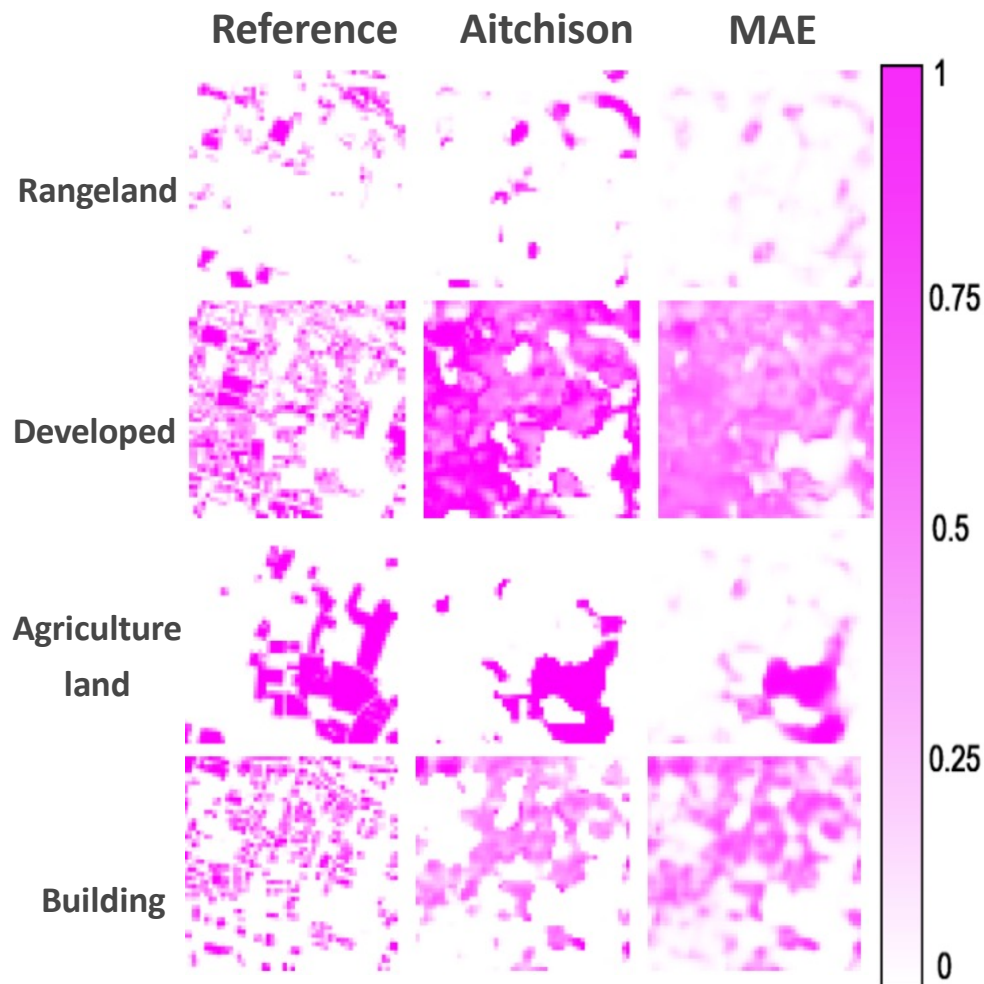
Aitchison loss for all models

Input	Model	MAE	RMSE	Aitchison Distance
Sentinel-2	MLP	0.118	0.246	9.773
	2D-CNN	0.115	0.198	9.667
	3D-CNN	0.112	0.187	9.498
Embedding V1	MLP	0.098	0.190	8.990
	2D-CNN	0.105	0.173	9.211
	3D-CNN	0.109	0.177	9.465



The optimal model architecture changed from 3D-CNN to MLP with Embedding V1.

② Effect of Loss Function



Class-wise Land-Cover Fraction Maps (Selected Classes)

Experimental Setting:

Embedding V1 → MLP

Aitchison loss produces
clearer distribution boundaries



**Aitchison loss produced realistic
land-cover fraction maps**

Effect of Input Features

Sentinel-2 Spectral Data

CNNs learn **spatial and spectral relationship** from image patches

⇒ CNN architectures outperform MLP

Embedding V1

Spatial, spectral, temporal, and semantic information are already **encoded**

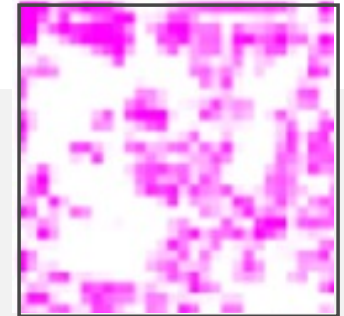
⇒ Additional feature extraction become **less important**

Effect of Loss Function

< Compositional nature of land cover fractions >

Most grid cells are dominated by **only a few classes**

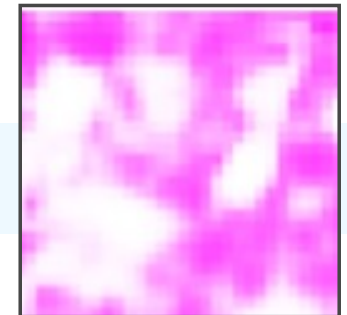
Many class fractions are **close to 0**



Regression loss (MAE)

Treats **all absolute differences equally**

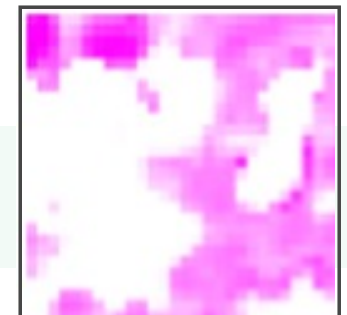
⇒ Pixels tend to be averaged



Aitchison loss

Evaluates **relative differences** between fractions.

⇒ Clearer land cover maps



- Introducing high-dimensional geospatial embeddings and a compositional loss function improved the accuracy of fractional land-cover mapping
- When using Google Satellite Embedding V1, the **simple MLP outperformed** more complex CNN-based models
- The Aitchison distance loss produced **more realistic and spatially coherent land-cover fraction maps**

Appendix: Training Settings

Optimizer	Adam
Learning Rate	1×10^{-3}
Epochs	200
Early Stopping	Patience = 20
Data Split	70% / 10% / 20% (Train / Validation / Test)
Kernel Size	3×3
Patch Size	5×5